# PHYS 4343 FS26: Atomic Physics

Exam 1

**Name:** _____

## Instructions

- **Time:** 60 minutes.
- This is a **closed-book and closed-notes** exam.
- The exam contains **three problems**. **Choose two of the three problems to solve.**
- Each problem is worth **50 points**. Your two selected problems therefore give a total of **100 points**.
- Clearly indicate which two problems you want to have graded. **Only two problems will be graded.**
- You do **not** receive additional credit for attempting all three problems.
- Show your reasoning and the important intermediate steps. An unsupported final answer may receive little or no credit.
- You may use a calculator.
- Any formulas or numerical information that are not expected to be known are provided in the individual problems.

**Read all three problems before deciding which two you want to solve.**

## Problem 1 — Hydrogen and Orbital Angular Momentum

Consider a hydrogen atom in the state

$$\psi(r,\vartheta,\varphi) = R_{32}(r)\left[\sqrt{\frac{1}{2}}\,Y_{22}(\vartheta,\varphi) + \frac{i}{2}\,Y_{20}(\vartheta,\varphi) - \frac{1}{2}\,Y_{2,-1}(\vartheta,\varphi)\right].$$

The functions $Y_{\ell m}$ are simultaneous eigenfunctions of $\hat{\vec{L}}^{\,2}$ and $\hat{L}_z$.

### (a) State vector — 5 points

Write the state in bra–ket notation using the basis states

$$|n,\ell,m\rangle\,.$$

### (b) Measurement of $\hat{\vec{L}}^{\,2}$ — 10 points

You measure the square of the orbital angular momentum,

$$\hat{\vec{L}}^{\,2}.$$

What result or results can be obtained, and with what probabilities?

### (c) Measurement of $\hat{L}_z$ — 15 points

Instead, you measure $\hat{L}_z$.

Determine

1. all possible measurement outcomes,
2. the probability of each outcome, and
3. the expectation value

$$\langle \hat{L}_z \rangle.$$

### (d) State after a measurement — 5 points

Suppose the measurement of $\hat{L}_z$ gives the result

$$L_z = 0.$$

What is the state of the atom immediately after the measurement?

**(e) Time evolution — 15 points**

Suppose the energies of the three states in the superposition are shifted to

$$E_{3,2,2} = 2\hbar\omega, \qquad E_{3,2,0} = 0, \qquad E_{3,2,-1} = -\hbar\omega.$$

Such energy shifts can, for example, arise from the interaction of the atom with a magnetic field.

1. Write the state $|\psi(t)\rangle$ at an arbitrary later time $t$.
   **10 points**

2. Does the spatial probability density $|\psi(\mathbf{r}, t)|^2$ remain constant in time? Briefly explain your answer.
   **5 points**

## Problem 2 — Angular Momentum Coupling

Consider an electron with orbital angular momentum $\ell = 1$ and spin $s = \frac{1}{2}$.

The total angular momentum is

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}.$$

**(a) Possible total angular momenta — 10 points**

What values of the total-angular-momentum quantum number $j$ are possible?

For each value of $j$, list the allowed values of $m_j$.

How many states are there altogether?

**(b) Coupling of orbital and spin angular momentum — 10 points**

Starting from

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}},$$

show that

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = \frac{1}{2}\left(\hat{\mathbf{J}}^2 - \hat{\mathbf{L}}^2 - \hat{\mathbf{S}}^2\right).$$

**(c) Spin-orbit interaction — 15 points**

Suppose the interaction between the orbital and spin angular momenta is described by

$$\hat{H}_0 = \frac{a}{\hbar^2}\hat{\mathbf{L}} \cdot \hat{\mathbf{S}},$$

where $a$ is a constant with units of energy.

Determine the energy for each possible value of $j$.

State the degeneracy of each energy level.

**(d) Magnetic-field splitting — 15 points**

Now suppose an additional interaction is present so that

$$\hat{H} = \frac{a}{\hbar^2}\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} + \frac{b}{\hbar}\hat{J}_z,$$

where $b$ also has units of energy.

Determine the energy eigenvalues in terms of $j$ and $m_j$.

List the energies of all states explicitly.

Which states are degenerate when $b = 0$? What happens to these degeneracies when $b \neq 0$?

## Problem 3 — The Proton Radius Puzzle

The **proton radius puzzle** concerns measurements of the proton charge radius. One way to determine the proton radius is through high-precision spectroscopy of hydrogen-like atoms. Atomic energy levels are slightly sensitive to the size of the proton because there is a small but nonzero probability of finding the orbiting particle inside the nucleus.

In 2010, a measurement using **muonic hydrogen** produced a proton-radius value that differed from previous measurements. Muonic hydrogen consists of a proton and a negatively charged muon, with

$$m_\mu = 207 m_e.$$

### (a) Ordinary hydrogen — 15 points

The normalized ground-state wavefunction of ordinary hydrogen is

$$\psi_{1s}(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0},$$

where

$$a_0 = \frac{4\pi\varepsilon_0 \hbar^2}{m_e e^2}.$$

Let $r_p$ denote the proton radius. Since

$$r_p \ll a_0,$$

you may assume that the wavefunction is approximately constant inside the proton,

$$\psi_{1s}(r) \approx \psi_{1s}(0) \qquad (r < r_p).$$

Estimate the probability of finding the electron inside the proton.

You may use

$$V = \frac{4}{3}\pi r_p^3.$$

### (b) Muonic hydrogen — 20 points

For a two-particle system,

$$\mu = \frac{m_1 m_2}{m_1 + m_2}.$$

Determine the reduced mass of muonic hydrogen in terms of $m_e$, using

$$m_p = 1836 m_e.$$

The hydrogen ground-state wavefunction can be adapted to muonic hydrogen by replacing the electron mass $m_e$ by the appropriate reduced mass.

Determine the corresponding characteristic length scale $a_\mu$ in terms of $a_0$.

Then write down the corresponding $1s$ wavefunction for muonic hydrogen and determine the probability of finding the muon inside the proton.

Again, you may assume that the wavefunction is approximately constant over the volume of the proton.

### (c) Physical interpretation — 15 points

Determine the ratio

$$\frac{P_\mu(r < r_p)}{P_e(r < r_p)}.$$

Explain briefly why muonic hydrogen is much more sensitive to the finite size of the proton than ordinary hydrogen.